TEXORA · LECTURE SHEET

# International Trade and the Theory of Comparative Advantage

*Why nations trade at all, the economic basis of trade, and the principle that explains it — comparative advantage — proved with a fully worked two-country example in which one country is better at producing everything, yet both still gain. Plus the gains from trade, the effect of quotas, and the Heckscher–Ohlin theorem.*

TE-4104 · APPAREL MERCHANDISING AND INTERNATIONAL TRADE · TOPIC 6 · LEVEL 4 TERM 1

## 1. Importance of international trade

**International trade** is the exchange of goods and services across national borders. Its importance, at principle level, is that it:

- **Widens choice and markets.** Consumers and firms obtain goods the home economy does not produce, or produces only at high cost; producers sell to a world market far larger than the home market.
- **Lets countries specialise.** Resources move towards the activities a country performs relatively best (Section 4), raising total output from the same resources.
- **Earns foreign exchange.** Export earnings pay for essential imports — fuel, machinery, raw materials — that the economy cannot supply itself.
- **Transfers technology and standards.** Trading firms adopt the quality, compliance and production standards of their markets (Topic 5), upgrading domestic capability.
- **Creates employment and income** in export industries and their supply chains, and disciplines domestic producers through competition.

## 2. Trend of international trade with reference to Bangladesh

Described qualitatively, as the course requires: Bangladesh's trade has grown from an import-dependent pattern dominated by a narrow range of primary exports towards an export structure led by manufactured goods, in which **readymade garments are the principal export sector** and the country is a major supplier to the apparel markets of Europe and North America. Knitwear and woven garments, home textiles and a smaller range of other manufactures (including leather goods, jute goods,

footwear and light engineering products, among others) make up the export basket, while imports are dominated by the inputs of that same industrial structure — textile raw materials and machinery — together with fuel, edible oils and other essentials. Two structural features define the trend at principle level: first, **concentration** — export earnings depend heavily on one sector, which is both the economy's strength and its exposure; second, **integration** — the apparel export sector itself depends on imported yarn, fabric and accessories for part of its production, so export growth and import growth move together. The remainder of this sheet supplies the theory that explains why such specialisation can benefit all trading partners.

## 3. Economic basis and sources of trade

Trade arises because countries differ. The principal sources of those differences are:

- **Resource endowments.** Countries hold different quantities of land, labour, capital and natural resources, so the same good costs different amounts of resources in different places.
- **Technology and productivity.** Knowledge, machinery and organisation make labour more productive in one country than another, industry by industry.
- **Climate and geography.** Some goods can only — or can most cheaply — be produced in particular conditions and locations.
- **Economies of scale.** Producing for a world market allows volumes no home market could absorb, lowering unit cost.
- **Product differentiation and taste.** Even similar economies trade varieties of the same product class to satisfy varied demand.

The common economic basis beneath all of these is a difference in **opportunity cost**: a good is imported when obtaining it through trade costs fewer domestic resources than producing it at home. The next section makes that precise.

## 4. The principle of comparative advantage

A country has an **absolute advantage** in a good if it can produce a unit with fewer resources than another country. It has a **comparative advantage** in the good for which its **opportunity cost** — what must be given up of the other good to produce one more unit — is lower. The principle of comparative advantage (associated with David Ricardo) states:

> **THE PRINCIPLE**

> Each country gains by specialising in the good in which it has a comparative advantage and trading for the other good — even if one country has an absolute advantage in both goods. What matters is relative, not absolute, efficiency.

### Worked example (illustrative, script-verified)

Two countries, A and B, each have **7,200 labour-hours**. Labour required per unit:

| COUNTRY | CLOTH (HOURS / UNIT) | YARN (HOURS / UNIT) | ABSOLUTE ADVANTAGE |
|---|---|---|---|
| A | 100 | 120 | — |
| B | 90 | 80 | B in both goods (90 < 100 and 80 < 120) |

**Step 1 — Opportunity costs.** Producing one unit of cloth uses hours that could have produced cloth-hours ÷ yarn-hours units of yarn:

Cloth: A = 100/120 = 0.83 units of yarn · B = 90/80 = 1.125 units of yarn → **A has the comparative advantage in cloth**

Yarn: B = 80/90 = 0.89 units of cloth · A = 120/100 = 1.20 units of cloth → **B has the comparative advantage in yarn**

**Step 2 — Autarky (no trade).** Each country divides its labour equally, 3,600 hours to each good:

| COUNTRY | CLOTH | YARN |
|---|---|---|
| A | 3,600 ÷ 100 = 36 | 3,600 ÷ 120 = 30 |
| B | 3,600 ÷ 90 = 40 | 3,600 ÷ 80 = 45 |
| **World total** | **76** | **75** |

**Step 3 — Complete specialisation** according to comparative advantage:

| COUNTRY | PRODUCTION |
|---|---|
| A — all labour to cloth | 7,200 ÷ 100 = **72 cloth** |
| B — all labour to yarn | 7,200 ÷ 80 = **90 yarn** |

**Step 4 — Trade at 1 cloth = 1 yarn** (a rate lying between the two opportunity costs of cloth, 0.83 and 1.125, so both sides accept it). A exports 30 cloth and imports 30 yarn; B exports 30 yarn and imports 30 cloth:

| COUNTRY | CONSUMPTION AFTER TRADE | AUTARKY CONSUMPTION | RESULT |
|---|---|---|---|
| A | 72 − 30 = **42 cloth** + **30 yarn** | 36 cloth + 30 yarn | Strictly better: +6 cloth, same yarn |
| B | **30 cloth** + 90 − 30 = **60 yarn** | 40 cloth + 45 yarn | A different, larger bundle: −10 cloth, +15 yarn — B is better off in value terms at the trade price (see note) |

**Note on B's bundle.** B consumes less cloth but much more yarn than in autarky. Valued at the trade price (1 cloth = 1 yarn), B's consumption is worth 30 + 60 = 90 units against autarky's 40 + 45 = 85 units, and B could have chosen any point on its trade line — including bundles with more of both goods than autarky for suitable exchange volumes. World output of the traded combination has risen from the same total labour: specialisation created the surplus that trade divides. A's gain is direct (+6 cloth for identical yarn); B's gain appears as a consumption possibility beyond its own production-possibility frontier, as Figure 2 shows.




Figure 1 — Opportunity-cost comparison for the worked example. Comparative advantage belongs to the lower bar in each pair — A for cloth, B for yarn — regardless of B's absolute advantage in both.

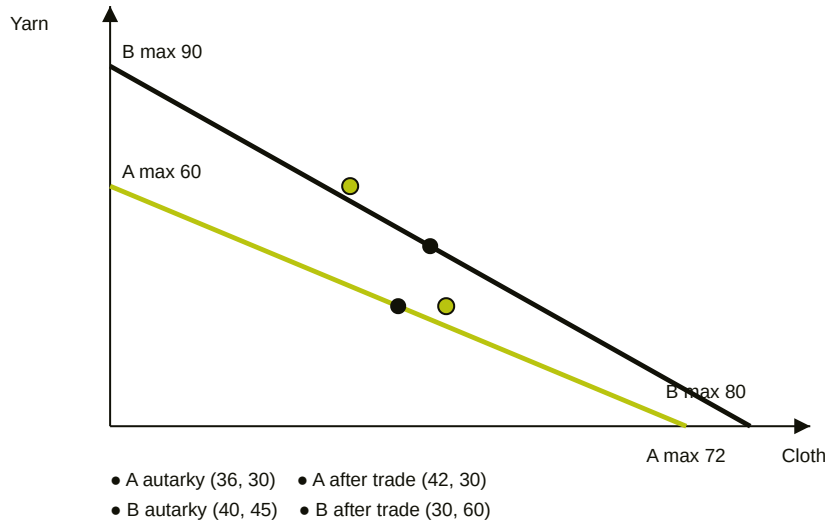



Figure 2 — PPF pair for the worked example (axes in units; A's frontier joins 72 cloth to 60 yarn, B's joins 80 cloth to 90 yarn). Both post-trade consumption points lie outside the producing country's own frontier — the visual form of the gains from trade.

## 5. Economic gains from trade

- **More output from the same resources** — specialisation by comparative advantage raises world production, as the example's trade bundles show.
- **Lower prices and wider choice** for consumers and for firms buying inputs.
- **Economies of scale** — export volumes support larger, more efficient production units.
- **Competition and innovation pressure**, and the inflow of technology, standards and foreign exchange described in Section 1.

The gains are aggregate: within a country, trade can shrink the sector in which the country lacks comparative advantage, so the distribution of gains — and assistance to adjusting workers and firms — is a real policy question (Topic 7). The existence of a national gain does not mean every individual gains automatically.

## 6. Effect of trade quotas

A **quota** is a direct limit on the quantity of a good that may be imported (or, in some arrangements, exported) in a period. Its effects, in principle:

- **Quantity is fixed, price adjusts.** Imports stop at the quota volume; the domestic price rises to whatever level clears the market at that restricted quantity — unlike a tariff, the government collects no revenue on the units that do enter.
- **Quota rents.** Whoever holds the right to import the limited quantity can buy at the world price and sell at the higher domestic price; the difference (the quota rent) goes to licence holders — domestic

importers or, under voluntary export restraints, to the foreign exporters.

- **Protection is certain in quantity, uncertain in price.** A quota guarantees domestic producers a maximum import volume, which is why industries seeking firm protection prefer it; a surge in demand raises price rather than imports.
- **Costs to consumers and efficiency** follow the price rise: less consumption, higher-cost domestic production expands, and resources are held in the protected activity instead of the comparative-advantage activity.
- For exporter countries, an allocated quota can also *guarantee market access* up to its volume — historically an important feature of apparel trade arrangements — while capping growth beyond it.

## 7. The Heckscher–Ohlin theorem

The **Heckscher–Ohlin theorem** (the syllabus book prints the name as "Heckscher-Ohilion", *sic*) explains comparative advantage by **factor endowments**. Its statement: a country exports the good that uses its **abundant factor** intensively, and imports the good that uses its **scarce factor** intensively. A labour-abundant country therefore tends to export labour-intensive goods — the principle-level explanation of a labour-abundant economy specialising in apparel, a labour-intensive manufacture — and to import capital-intensive goods such as machinery. Where Ricardo's example traced advantage to productivity differences, Heckscher–Ohlin traces it to the relative abundance of labour, capital and land, and predicts that trade tends to equalise factor returns between countries over time. Together the two theories answer the merchandiser's background question — why the orders are here — at the level of the whole economy.

## Glossary

| TERM | MEANING |
|---|---|
| Absolute advantage | Producing a unit of a good with fewer resources than another country. |
| Comparative advantage | Producing a good at lower opportunity cost than another country. |
| Opportunity cost | The amount of another good given up to produce one more unit of a good. |
| Autarky | The no-trade position: a country consumes only what it produces. |
| Production-possibility frontier | The boundary of output combinations a country can produce with its resources. |
| Quota | A direct quantitative limit on imports (or exports) in a period. |

| | |
|---|---|
| Heckscher–Ohlin theorem | The theory that countries export goods intensive in their abundant factor and import goods intensive in their scarce factor. |

## Key points and summary

- Trade's basis is difference — in endowments, technology, climate and scale — expressed as differences in opportunity cost.
- Comparative advantage, not absolute advantage, decides the profitable pattern of specialisation.
- Worked example: B is absolutely better at both goods, yet A specialises in cloth (opportunity cost 0.83 < 1.125) and B in yarn (0.89 < 1.20); trading at 1 : 1 leaves A consuming 42 cloth + 30 yarn against autarky's 36 + 30 — strictly better — and B on a consumption point beyond its own frontier.
- Gains from trade are real but aggregate; quotas fix import quantity, raise domestic price, and create rents for licence holders instead of revenue for government.
- Heckscher–Ohlin grounds comparative advantage in factor abundance: labour-abundant economies export labour-intensive goods.
- Bangladesh's trade trend, at principle level: manufactured, apparel-led exports integrated with imported inputs — concentration as both strength and exposure.

## Practice

Answer in your own words; brief model answers follow. Show all working for numerical questions.

1. Country C needs 50 hours per unit of cloth and 100 hours per unit of yarn; country D needs 60 hours for cloth and 90 for yarn. Compute all four opportunity costs and state each country's comparative advantage.
2. In the worked example, verify that A's post-trade consumption point (42 cloth, 30 yarn) lies beyond A's own PPF.
3. State two ways in which a quota differs from a tariff in its effects.
4. Using Heckscher–Ohlin, explain in two sentences why a labour-abundant country is expected to export apparel and import textile machinery.

### Answer key

1. Cloth: C = 50/100 = 0.50 yarn; D = 60/90 = 0.67 yarn → C has CA in cloth. Yarn: D = 90/60 = 1.50 cloth; C = 100/50 = 2.00 cloth → D has CA in yarn. (D has absolute advantage in yarn only; C in cloth only — CA follows the opportunity costs.)
2. Producing 42 cloth and 30 yarn itself would cost A 42 × 100 + 30 × 120 = 4,200 + 3,600 = 7,800 hours, but A has only 7,200 hours — the bundle is unattainable without trade, i.e. beyond its PPF.

3. A tariff fixes the tax and lets quantity adjust, collecting government revenue; a quota fixes quantity and lets price adjust, creating rents for licence holders and no revenue on the units admitted.
4. Apparel production uses labour intensively and labour is the abundant (hence relatively cheap) factor, so apparel is the country’s low-opportunity-cost export. Machinery uses capital intensively and capital is the scarce factor, so machinery is produced more cheaply abroad and imported.

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